FEASIBILITY STUDY AND IMPLEMENTATION OF EV CONVERSIONS FOR MOTORCYCLES IN SRI LANKA

FINAL YEAR PROJECT

CALCULATIONS

DATA FOR CALCULATION

* Considering the CT100 bike model,

- Gross weight → 241 kg
- Width → 752 mm
- Height → 1071 mm
- Wheel length → 1965 mm ($2\pi r$), Linear distance
- Single rider → 68 kg
- C_r → 0.004 (for asphalt road)
  (Coefficient of rolling resistance)
- C_A → 0.88 (for bike)
  (Coefficient of air resistance)
- ρ → 1.23 kg/m³
- V → 60 km/h = 16.67 m/s
- Efficiency → 85%

MOTOR SELECTION

Every vehicle should overcome these resistance forces,

- Rolling resistance (F_R)
- Gradient resistance (F_G)
- Aerodynamic resistance (F_A)

Therefore, Total resistance force can be get,

$$F_{Total} = F_R + F_G + F_A \quad \text{--- (1)}$$

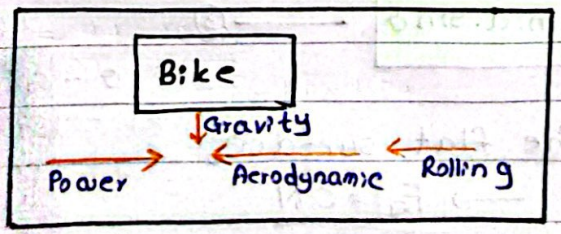

Figure 1 : Free body diagram

Rolling resistance (F_R)

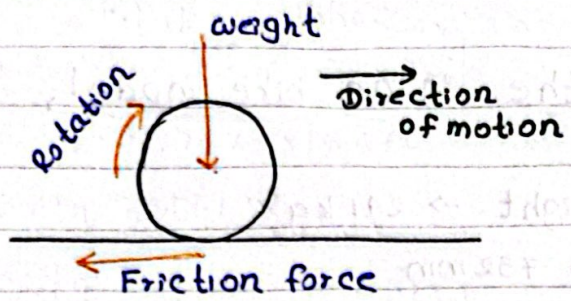

Figure 2 : FBD for rolling

$$F_{rolling} = F_R = C_r \times m \times a \quad \text{——— (2)}$$

C_r - Coefficient of rolling resistance

m - Mass of the vehicle (kg)

a - Acceleration due to gravity (m/s^2)

* Using equation (2),

$$F_R = 0.004 \times (241 + 68)\,kg \times 9.81\,m/s^2$$

$$= \underline{\underline{12.125\,N}}$$

Gradient resistance (F_G)

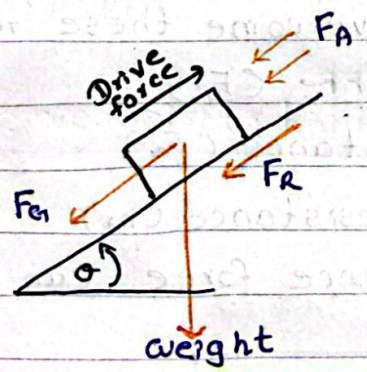

Figure 3 : FBD for gradient

$$F_G = m.a.\sin\theta \quad \text{——— (3)}$$

θ - Gradient angle.

Considering the flat surface,

$\theta = 0^\circ \longrightarrow F_G = \underline{0\,N}$

Aerodynamic resistance

$$F_A = 0.5\,(\rho \times V^2 \times C_A \times A_f) \quad \text{—— (4)}$$

ρ - Density of air (kg/m^3)
V - Velocity of vehicle (m/s)
C_A - Coefficient of air resistance
A_f - Frontal area of vehicle (m^2)

⇒ A_f calculation

* For motorcycle,
  - width → Up to length of handbar
  - Height → Up to person's helmet

$$A_f = \text{Height} \times \text{width} \times \text{Adjusting value} \quad \text{—— (5)}$$

Vehicle	Adjusting value
Cars	85%
Bikes	70%
Trucks / buses	100%

Table 1 : Adjusting value for different vehicles

Using equation (5),

$$A_f = (1.071 \times 0.752) \times 0.70$$
$$= 0.564\ m^2$$

Using equation (4),

$$F_A = 0.5 \times 1.23 \times (16.67)^2 \times 0.88 \times 0.564$$
$$= \underline{84.822\,N}$$

Using equation (1),

$$F_{Total} = F_R + F_G + F_A$$
$$= (12.125 + 0 + 84.822)\,N$$
$$= \underline{96.947\,N}$$

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Therefore,

$$\text{Power} = \text{Total resistance force} \times \text{Velocity}$$

$$= 96.947 \times 16.67$$

$$= \underline{\underline{1616.1\ W}}$$

$$\text{RPM} = \frac{\text{Total distance covered per hour}}{\text{Linear distance}}$$

$$= \frac{60000}{1.965 \times 60}$$

$$= 508.9\ \text{rpm} \simeq 509\ \text{rpm}$$

⇒ <u>Torque</u> (τ)

$$\eta = \frac{P_{out}}{P_{in}}$$

$$P_{out} = \eta \times P_{in}$$

$$\tau \times \omega = 0.85 \times 1616.1$$

$$\tau = \frac{0.85 \times 1616.1 \times 60}{2\pi \times 509}$$

$$= 25.771\ Nm$$

$$r = \frac{1.965}{2\pi} = 0.313\ m$$

$$\tau = \text{radius} \times \text{force}$$

$$= 0.313 \times 96.947$$

$$= 30.344\ Nm$$

* Therefore, we selected 48V, 1800W motor.

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BATTERY SELECTION

$$\text{Current} = \frac{1800W}{48V} = 37.5 \text{ Amps}$$

(Theoretically)

* Now we find out the watt hour of the battery.

⊙ To run the 1800W for 1 hour,

$= 1800W \times 1 \text{ hour}$

$= 1800$ watt hour

⊙ Take efficiency of 80% for battery,

$$= \frac{1800}{0.8} = 2250 \text{ watt hour}$$

* Convert watt hour of battery into Amphere hour of battery.

Power = Voltage × Current
watt hour = Voltage × Amphere hour

$2250 = 48 \times \text{Amphere hour}$

$$\text{Amphere hour} = \frac{2250}{48}$$

$= 46.875$ Ah

* We selected 48V 60Ah battery pack.

⇒ Special points

* If we use lead acid
  ↳ ~~33A~~ $46.875 \times 1.8 = 84.375$ Ah
* If we use Li polymer
  ↳ $46.875 \times 1.05 = 49.218$ Ah

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